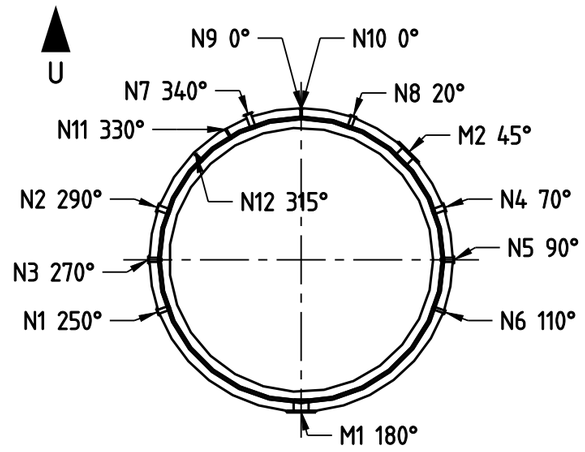
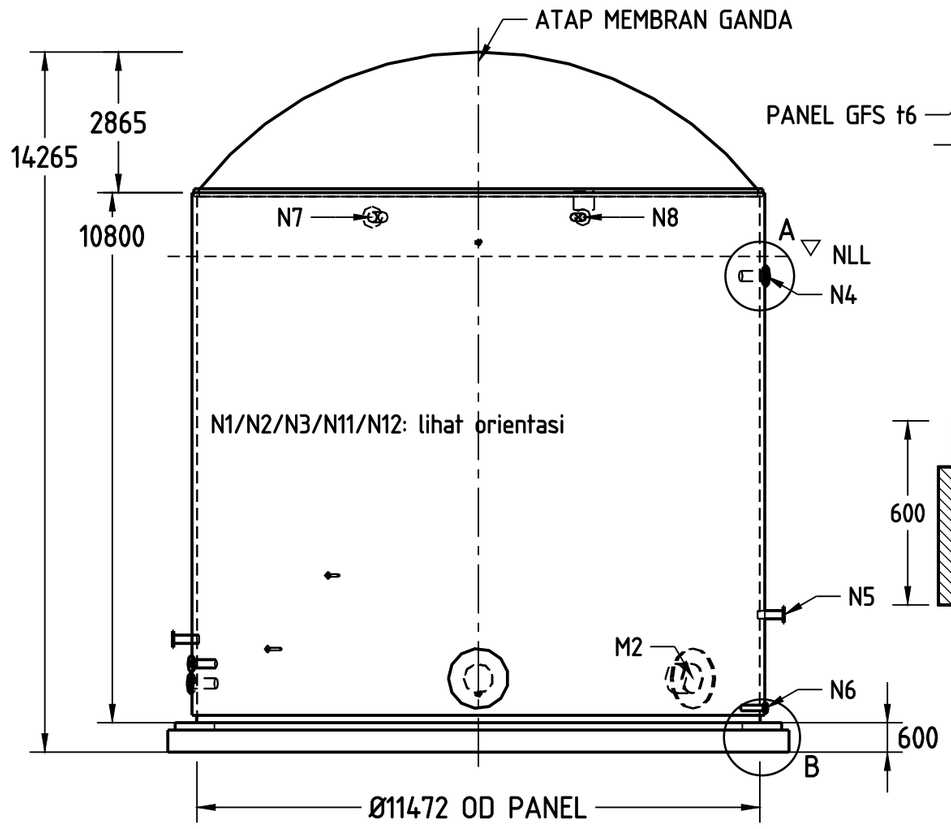
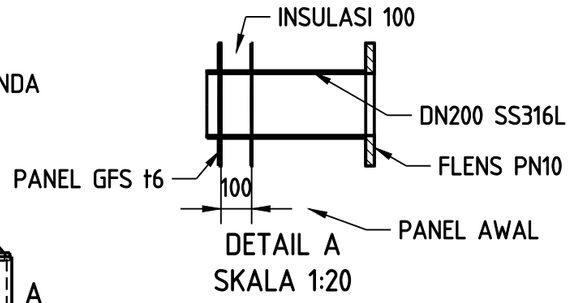


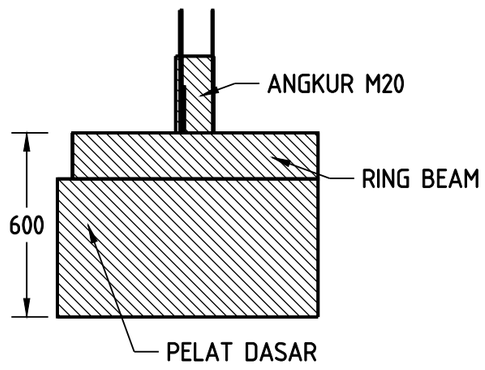

## ORIENTASI NOSEL
SKALA 1:250




## ELEVASI (DARI SELATAN)
SKALA 1:125




DETAIL A
SKALA 1:20



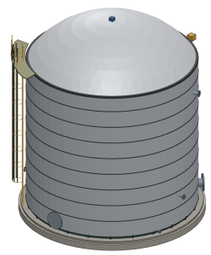

DETAIL B
SKALA 1:20

| JADWAL NOSEL | | | | | |
|---|---|---|---|---|---|
| Tag | Ukuran | Rating | Layanan | Elev. | Orient. |
| N1 | DN150 | PN10 RF | Umpan reaktor (dari P-201) | 1800 | 250° |
| N2 | DN200 | PN10 RF | Isap jet mixing (ke P-202) | 1400 | 290° |
| N3 | DN150 | PN10 RF | Balik jet mixing (header nosel jet) | 2300 | 270° |
| N4 | DN200 | PN10 RF | Keluaran ke degasser DG-301 (luapan) | 9700 | 70° |
| N5 | DN150 | PN10 RF | Resirkulasi lumpur dari LS-301 | 2800 | 90° |
| N6 | DN100 | PN10 RF | Pengurasan dasar / pasir | 900 | 110° |
| N7 | DN200 | PN10 RF | Keluaran biogas (ruang gas) | 10900 | 340° |
| N8 | DN150 | PN10 RF | PVRV tekanan-vakum | 10900 | 20° |
| N9 | DN50 | PN10 RF | Pemancar level LT atas | 10400 | 0° |
| N10 | DN50 | PN10 RF | Pemancar level LT bawah | 1200 | 0° |
| N11 | DN50 | PN10 RF | Termometer TT / pH | 3600 | 330° |
| N12 | DN50 | PN10 RF | Titik sampel | 2100 | 315° |
| M1 | DN600 | PN10 RF | Lubang orang dinding (bawah) | 1500 | 180° |
| M2 | DN600 | PN10 RF | Lubang orang dinding (seberang) | 1500 | 45° |

| DATA DESAIN | |
|---|---|
| Data desain | Nilai |
| Fungsi / jumlah | Reaktor kontak termofilik, 3 unit identik (2 / 4 untuk 30 / 60 t/jam) |
| Kode desain | AWWA D103-09 (tangki baja dibaut) + ISO 28765 (enamel) |
| Tekanan / suhu desain | +50 / −5 mbar(g) + hidrostatik penuh; 70 °C (op. 55 °C) |
| Volume dinding / kerja | 1114 m³ / 980 m³, NLL 9500 mm dari lantai |
| Panel | 126 panel 2700 x 1200 mm, t 5–10 mm (rata-rata 6), enamel 2 lapis |
| Atap | Membran ganda PVC-PES (dalam: gas, luar: udara), tinggi 2865 mm |
| Angin / gempa | SNI 1727:2020 (40 m/s) / SNI 1726:2019, Kelas D |
| Insulasi / kladding | Mineral wool 100 mm 100 kg/m³ / Al 0,8 mm + sabuk |
| Massa baja / operasi | ± 20.3 t / ± 1020 t |
| Pengiriman / relokasi | Panel datar dalam 2 kontainer 40 ft; ereksi ± 3 minggu; dapat dibongkar |

| DAFTAR MATERIAL (MASSA DARI MODEL 3D) | | | | |
|---|---|---|---|---|
| No | Bagian | Material | Jml | Massa kg |
| 1 | Panel dinding GFS | S355 + enamel 2 lapis | 126 | 18663 |
| 2 | Baut panel + sealant | M12 galv. + mastik | ± 7560 | – |
| 3 | Sudut penahan + angkur M20 | S275 galv. | 24 | 452 |
| 4 | Leher nosel + flens | SS316L / SA-105 | 14 | 1667 |
| 5 | Header jet internal + 8 nosel | SS316L DN150 sch10 | 25.6 m | 282 |
| 6 | Atap membran ganda + klem | PVC-PES 1.100 g/m² / Al | 1 | ± 950 |
| 7 | Blower udara membran luar | 0,75 kW + katup cek | 1 | 60 |
| 8 | Tangga + sangkar + platform | S275 galv., grating | 1 | 1267 |
| 9 | Insulasi mineral wool | 100 kg/m³ | 39.3 m³ | 3926 |
| 10 | Kladding + sabuk | Al 3003 t0,8 | 1 | 1203 |
| 11 | Ring beam + pelat dasar | Beton K-300 (sipil lokal) | 1 | – |

CATATAN :
1. Panel dibaut dengan sealant; tanpa las di lokasi.
2. Uji: isi air 24 jam (kebocoran), uji udara atap 5 mbar.
3. Elevasi dari muka tanah (EL 0); orientasi dari utara.
4. Relokasi: bongkar panel, ganti sealant & ring beam.



ISOMETRI

| | | | | |
|---|---|---|---|---|
| University: UNIVERSITAS | Checked by : – | Design Lab. | Document type: Equipment Drawing | Rev: 2 |
| | Author : NAMA TIM | Reaktor Kontak Termofilik R-201 s/d R-203 (GFS) | Unit: mm / Scale: 1:125/1:250/1:20 / Creation date: 15/10/2026 | Size: A3 |
| | Eng. Approved by : – | | BG-V-201 | Sheet number: 2 / 9 |